

## Problem

Use the Laplace transform to solve the integrodifferential equation

$$\frac{dy}{dt} + 3y(t) + 2 \int_0^t y(\tau) d\tau = 2e^{-3t}, \quad y(0) = 0$$

## Step-by-step solution

## Step 1 of 3

Given integrodifferential equation is

$$\frac{dy(t)}{dt} + 3y(t) + 2 \int_0^t y(\tau) d\tau = 2e^{-3t}$$

Taking transform of each term, we get

$$[sY(s) - y(0)] + 3Y(s) + \frac{2}{s}Y(s) = \frac{2}{s+3}$$

Substitute,  $y(0) = 0$  we obtain

$$sY(s) + 3Y(s) + \frac{2}{s}Y(s) = \frac{2}{s+3}$$

Multiplying  $s$  on both sides

$$s^2Y(s) + 3sY(s) + 2Y(s) = \frac{2s}{s+3}$$

$$(s^2 + 3s + 2)Y(s) = \frac{2s}{s+3}$$

$$Y(s) = \frac{2s}{(s+1)(s+2)(s+3)}$$

## Step 2 of 3

Using partial expansion

$$\frac{2s}{(s+1)(s+2)(s+3)} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3} \quad \text{----- (1)}$$

Multiplying  $(s+1)(s+2)(s+3)$  on both sides

$$2s = A(s+2)(s+3) + B(s+1)(s+3) + C(s+1)(s+2)$$

Consider  $s = -1$

$$A(1)(2) = -2$$

$$A = -1$$

Consider  $s = -2$

$$B(-1)(1) = -4$$

$$B = 4$$

Consider  $s = -3$

$$C(-2)(-1) = -6$$

$$C = -3$$

Substituting  $A = -1$ ,  $B = 4$  and  $C = -3$  in equation (1)

$$Y(s) = \frac{-1}{s+1} + \frac{4}{s+2} - \frac{3}{s+3}$$

Taking inverse transform on both sides

$$\therefore y(t) = (-e^{-t} + 4e^{-2t} - 3e^{-3t})u(t)$$